Lab 1 – Part 4: Comparing TCP and UDP

Task 1

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP opens a connection with a three-way handshake and resends lost packets. UDP does neither.
Data Integrity and Ordering	TCP	TCP uses sequence numbers and ACKs to keep data in order and resend missing parts. UDP only detects errors with a checksum.

Task 2: Use cases and Performance

	TCP	UDP
Use cases	Web browsing (HTTP/HTTPS), email, file transfer, remote login (SSH).	Video streaming, video calls, online gaming, DNS.
Performance	Slower: bigger header (20 bytes or more), a handshake before sending data, and waiting for ACKs. But data always arrives complete and in order.	Faster: small 8-byte header, no handshake, no ACKs. But lost packets are not recovered.